(Chapter 14); with Wald, gave the first argument suggesting that time machines might be destroyed whenever they try to form (Chapter 14).

Giacconi, Riccardo (b. 1931). Italian born, American experimental physicist and astrophysicist; led the team that discovered the first X-ray star, in 1962, using a detector flown on a rocket (Chapter 8); led the team that designed and built the Uhuru X-ray satellite, which produced the first strong X-ray evidence that Cygnus X-1 is a black hole (Chapter 8).

Ginzburg, Vitaly Lazarevich (b. 1916). Soviet theoretical physicist; invented the LiD fuel for the Soviet hydrogen bomb and then was separated from the bomb project (Chapter 6); with Landau, developed an explanation for the origin of superconductivity (Chapters 6 and 9); discovered the first evidence that a black hole has no hair (Chapter 7); developed the synchrotron radiation explanation for the origin of cosmic radio waves (Chapter 9).

Greenstein, Jesse L. (b. 1909). American optical astronomer; colleague of Zwicky's (Chapter 5); with Fred Whipple found it impossible to explain cosmic radio waves (Chapter 9); triggered the beginning of America's research effort in radio astronomy (Chapter 9); with Maarten Schmidt, discovered quasars (Chapter 9).

Hartle, James B. (b. 1939). Student of Wheeler's; with Bekenstein, showed that one cannot discern, by any external study of a black hole, what kinds of particles were among the material that formed it (Chapter 7); with Hawking, discovered the laws that govern the evolution of a black hole's horizon (Chapter 12); with Hawking, is developing insights into the laws of quantum gravity (Chapter 13).

Hawking, Stephen W. (b. 1942). British theoretical physicist; student of Sciama's; developed key parts of the proof that a black hole has no hair (Chapter 7); with Bardeen and Carter, discovered the four laws of black-hole mechanics (the laws of evolution of black holes) (Chapter 12); discovered that, if one ignores the laws of quantum mechanics, the surface areas of black holes can only increase, but quantum mechanics makes black holes evaporate and shrink (Chapter 12); showed that tiny black holes could have formed in the big bang and, with Page, placed observational limits on such primordial holes based on astronomers not seeing gamma rays produced by their evaporation (Chapter 12); developed global (topological) methods for analyzing black holes (Chapter 13); with Penrose, proved that the big bang contained a singularity (Chapter 13); formulated the